LUCIDMSK CME PROGRAM

MODULE 09

Elbow MRI: Ligaments, Nerves, OCD & Tendon Injuries

UCL Grading · Capitellar OCD · Lateral Epicondylosis · PLRI · Ulnar Nerve · Distal Biceps

1.0 AMA PRA Cat 1 | **Elbow** | **CME Article Review**

LEARNING OBJECTIVES

1. Grade UCL (medial collateral ligament) injuries on coronal MRI.
2. Distinguish stable from unstable OCD lesions of the capitellum.
3. Identify lateral epicondylosis (ECRB) and LUCL tears on MRI.
4. Recognize PLRI and its relationship to LUCL disruption.
5. Interpret ulnar nerve CSA and identify cubital tunnel syndrome on MRI.
6. Diagnose distal biceps and triceps tendon tears on axial and sagittal MRI.

1. UCL (Medial Collateral Ligament) Grading

The anterior bundle of the UCL is the primary valgus stabilizer. Assess on coronal PD fat-sat. Most tears occur proximally at the medial epicondyle in overhead throwing athletes.

UCL Grading — Coronal PD / T2 Fat-Sat

Grade	Pathology	MRI Findings	Management
I — Sprain	Intact fibers	Periligamentous edema; thickened; continuous fibers	Conservative; rehabilitation
II — Partial	Partial fiber disruption	Partial discontinuity; T-sign (undersurface); T2 signal within	Conservative vs. PRP vs. surgery (throwers)
III — Complete	Full-thickness rupture	Complete discontinuity; fluid gap; often proximal	UCL reconstruction (Tommy John) for overhead athletes

Additional UCL Measurements

Measurement	Normal Value	Clinical Use
UCL thickness (coronal)	≤3–4 mm	Thickening >5 mm = chronic tear/scar tissue
Medial joint space	2–4 mm	Widening >4 mm suggests medial laxity/instability
Common flexor origin signal	Homogeneous low T1/T2	T2 signal = medial epicondylosis (golfer's elbow)

2. Capitellar OCD — Stability Criteria

OCD Stability — MRI Criteria

Finding	Stable	Unstable
Overlying cartilage	Intact; homogeneous	Breached; defect visible
T2 beneath fragment	None (no undercutting)	HIGH T2 fluid UNDERCUTTING fragment (key unstable sign)
Fragment position	In situ; no displacement	Loose or partially detached
Rim enhancement (post-Gd)	Present (granulation = healing)	Absent/thin = fibrocartilage, not healing
Loose body	Absent	Present (displaced fragment)

KEY POINT

Single most important unstable criterion: T2 high-signal fluid UNDERCUTTING the osteochondral fragment. If present → unstable → arthroscopic fixation or removal. Stable (intact cartilage, no undercutting) → non-operative with activity restriction + serial MRI.

3. Lateral Elbow, Ulnar Nerve & Distal Biceps

Lateral Elbow Structures — MRI Review

Structure	Pathology	Key MRI Finding
ECRB (common extensor origin)	Lateral epicondylosis (tennis elbow)	T2 signal at origin; partial or complete tear
LUCL	PLRI when disrupted	Discontinuity on coronal; pivot-shift test positive
Radial collateral ligament	Varus instability (rare isolated)	Coronal MRI; less often injured alone

Ulnar Nerve Assessment at Cubital Tunnel (Axial MRI)

Parameter	Normal	Abnormal
CSA at cubital tunnel	≤9–10 mm²	> 10 mm² = enlargement; cubital tunnel syndrome
Intrinsic T2 signal	Homogeneous; matches surrounding	Increased T2 = edema/demyelination
Flexor carpi ulnaris	Normal signal/bulk	T2 high signal = acute denervation edema
Nerve position	Stable in groove	Subluxation = snapping ulnar nerve

Distal Biceps Tear Classification (Axial MRI at Radial Tuberosity)

Type	Finding	Key Clue
Tendinopathy	Thickened; intratendinous T2; no gap	Fusiform T2 at tuberosity; posterior fibers prone
Partial tear	Partial fiber discontinuity	Posterior (bursal) surface most common
Complete tear	Full-thickness gap; not visible at tuberosity	Empty tuberosity on axial; proximal retraction on sagittal

TEACHING PEARL

Distal biceps: axial plane at the radial tuberosity is primary. Complete tears show "empty tuberosity" on axial. Report retraction level on sagittal for surgical planning. Most partial tears affect short-head/posterior fibers.

References

- Potter HG et al. Clin Sports Med 2004.
- Badia A et al. J Hand Surg 2006.